WORKED SOLUTION

STEP III 2004, Question 14

Probability and Statistics • Hypothesis testing

STEP III 2004, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q14

The key idea

Both hypotheses give $\overline{X}$ the same centre μ; they differ only in how spread out $\overline{X}$ is. So the evidence against H_0 is $\overline{X}$ landing *far from* μ, on either side, and that is exactly what a symmetric two-sided rejection region detects. After that, everything reduces to standardising $\overline{X}$, once under each hypothesis.

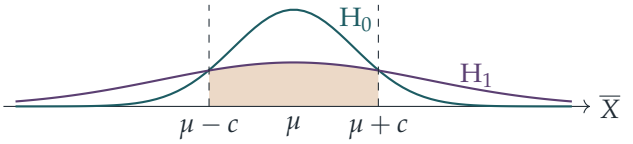
Solution

Why a two-sided test

Since $X \sim N(\mu, \sigma^2)$ and the sample values are independent, the sample mean satisfies

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Under either hypothesis $\overline{X}$ is centred at the known value μ; the only difference is that under H_1 its standard deviation $\sigma_1/\sqrt{n}$ is larger than $\sigma_0/\sqrt{n}$. Values of $\overline{X}$ close to μ are relatively more likely under H_0, while values far from μ are relatively more likely under H_1 (the wider curve has the fatter tails; see the figure). A large value of $\overline{X}$ is no more evidence for H_1 than an equally small one, because both distributions are symmetric about μ. So the natural test rejects H_0 when $|\overline{X} - \mu|$ is large, in either direction, and by symmetry the acceptance interval should be centred on μ: accept H_0 when $\mu - c < \overline{X} < \mu + c$. A one-sided test would be wrong here, as it would ignore half of the evidence for the larger spread.



The shaded area (accepting H_0 although H_1 is true) is β; the teal area outside the dashed

lines is α.

Finding c

Under H_0, $Z = \dfrac{\sqrt{n}\,(\overline{X} - \mu)}{\sigma_0}$ is standard normal. We accept H_1 when $|\overline{X} - \mu| \geq c$, that is when $|Z| \geq c\sqrt{n}/\sigma_0$. By the symmetry of the standard normal,

$$\alpha = \mathrm{P}\left(|Z| \geq \frac{c\sqrt{n}}{\sigma_0}\right) = 2\left(1 - \Phi\left(\frac{c\sqrt{n}}{\sigma_0}\right)\right), \qquad \text{so} \qquad \Phi\left(\frac{c\sqrt{n}}{\sigma_0}\right) = 1 - \tfrac{1}{2}\alpha.$$

Since Φ is strictly increasing, this says $c\sqrt{n}/\sigma_0 = z_\alpha$, and therefore

$$c = \frac{z_\alpha\,\sigma_0}{\sqrt{n}}.$$

β does not depend on n

Under H_1, $W = \dfrac{\sqrt{n}\,(\overline{X} - \mu)}{\sigma_1}$ is standard normal, and

$$\begin{aligned}\beta = \mathrm{P}(|\overline{X} - \mu| < c) &= \mathrm{P}\left(|W| < \frac{c\sqrt{n}}{\sigma_1}\right) = \mathrm{P}\left(|W| < \frac{z_\alpha \sigma_0}{\sigma_1}\right) \\ &= 2\,\Phi\left(\frac{z_\alpha \sigma_0}{\sigma_1}\right) - 1,\end{aligned}$$

where we substituted $c\sqrt{n} = z_\alpha \sigma_0$ from the previous part. The $\sqrt{n}$ in c cancels the $\sqrt{n}$ from standardising, so β depends only on α and σ_0/σ_1, not on n.

> **Ask yourself** Surely a bigger sample should make the test better?

Usually it does, because the two sampling distributions pull apart as n grows. Here they never do: both are centred at μ, and increasing n shrinks *both* standard deviations by the same factor $\sqrt{n}$. The whole picture simply rescales, so the areas α and β are unchanged. The sample mean is a poor statistic for testing a variance; it throws away the information about spread that the individual values contain.

The minimum ratio σ_1/σ_0

Write $r = \sigma_1/\sigma_0 > 1$.

Condition $\alpha < 0.05$. This means $\Phi(z_\alpha) = 1 - \frac{1}{2}\alpha > 0.975 \approx \Phi(1.960)$, so, as Φ is increasing, $z_\alpha > 1.960$.

Condition $\beta < 0.05$. This means $2\Phi(z_\alpha/r) - 1 < 0.05$, that is $\Phi(z_\alpha/r) < 0.525 \approx \Phi(0.063)$, so $z_\alpha/r < 0.063$, i.e.

$$r > \frac{z_\alpha}{0.063}.$$

Combining, we need $r > z_\alpha/0.063 > 1.960/0.063$. Conversely, if $r > 1.960/0.063$ we can choose z_α just above 1.960 with $z_\alpha < 0.063\,r$, and then both conditions hold. Hence

the ratio must exceed

$$\frac{1.960}{0.063} \approx 31.1,$$

so the minimum value of σ_1/σ_0 is approximately 31. The standard deviations must differ by a factor of about thirty before the sample mean can tell them apart this reliably, whatever the sample size.

Checks

- A simulation with $\mu = 3$, $\sigma_0 = 1$, $\sigma_1 = 2.5$, $\alpha = 0.1$ and $n = 1, 4, 9, 25$ gave rejection rates under H_0 of about 0.10 every time, and acceptance rates under H_1 of about 0.489 for every n, matching $2\Phi(1.645/2.5) - 1 \approx 0.489$.
- With $r = 1.960/0.063$ and $\alpha = 0.05$ exactly, $\beta = 2\Phi(0.063) - 1 \approx 0.050$, on the boundary as expected. Using exact inverse values of Φ gives $r \approx 31.3$, so the answer 31 is robust to the rounding in the given table values.
- Sanity: if $\sigma_1 = \sigma_0$ then $\beta = 2\Phi(z_\alpha) - 1 = 1 - \alpha$, as it must be when the hypotheses coincide.

Common slips

- Using a one-sided region such as $\overline{X} > \mu + c$: under H_1 the mean doesn't move, so there is no reason to favour one tail.
- Writing $\alpha = 1 - \Phi(c\sqrt{n}/\sigma_0)$, forgetting that the rejection region has two tails.
- Getting the inequality for r the wrong way round: small β needs z_α/r *small*, so r large.

Beyond STEP

This question shows the trade-off between the two error types and why the choice of test statistic matters. The Neyman–Pearson lemma says the most powerful test compares the likelihoods under the two hypotheses; here that leads to the statistic $\sum(X_i - \mu)^2$, whose χ^2_n distribution does separate the hypotheses as n grows, unlike $\overline{X}$. For a readable introduction to these ideas, see Blitzstein and Hwang, *Introduction to Probability*, or Grimmett and Welsh, *Probability: An Introduction*.

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